Math 133 Written Homework 7

Differentiation, Extrema, and Concavity

Due Thursday, October 29, 2026, at 11:59 p.m.

Show your work, including the differentiation rules you use and reasons for your conclusions. Justify classifications of critical points and inflection points using derivative signs or appropriate derivative tests. Give exact values when possible, and include units in business applications. Label axes and important points on your graphs.

List collaborators, time spent, and any use of generative AI with your submission. Your submitted work must be your own. Late work will not be accepted.

1. Differentiate each function using the product or quotient rule.

   (a) $h(x) = (x^2 + 1)e^{-x}$.

   (b) $k(x) = \dfrac{\ln x}{x + 2}$.

2. Differentiate each function, and state where its derivative is defined.

   (a) $u(x) = e^{x^2 - 3x}$.

   (b) $v(x) = 5^{2x+1}$.

   (c) $w(x) = \ln(x^2 + 4)$.

   (d) $z(x) = \log_2(3x - 1)$.

3. Let $f(x) = x^3 - 6x^2 + 9x + 2$. Find all critical numbers and classify each as a local maximum, a local minimum, or neither. Determine the intervals where f is increasing or decreasing and where it is concave up or concave down. Find all inflection points, and sketch the graph using your conclusions.

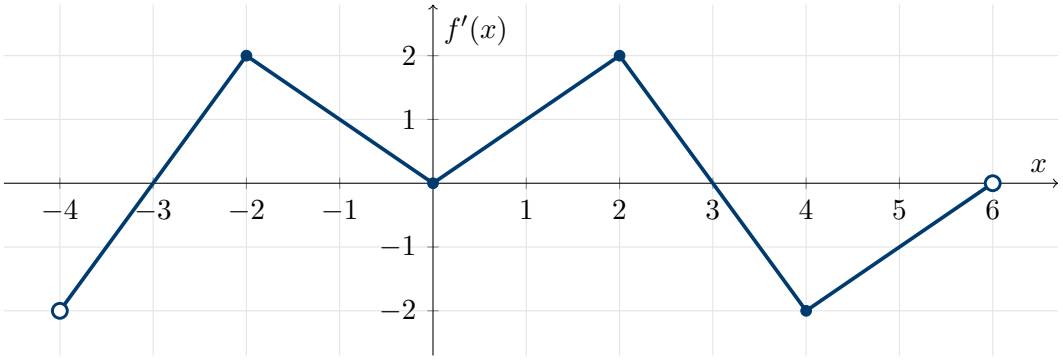
4. Let $g(x) = x - 2\ln x$. State its domain, find all critical numbers, and classify them. Determine the intervals where g is increasing or decreasing and where it is concave up or concave down. Does g have any inflection points?

5. A company sells q items at a price of

$$p(q) = 60e^{-q/100}, \qquad q \geq 0,$$

in dollars per item. Its total revenue is $R(q) = qp(q)$. Find $R'(q)$ and $R''(q)$. Find and classify all critical numbers of R for $q > 0$. At what quantity is revenue maximized? Determine where revenue is concave up or concave down, and find the quantity at which its concavity changes.

6. The graph below shows f' for a differentiable function f on $(-4, 6)$. The segments shown are straight lines.

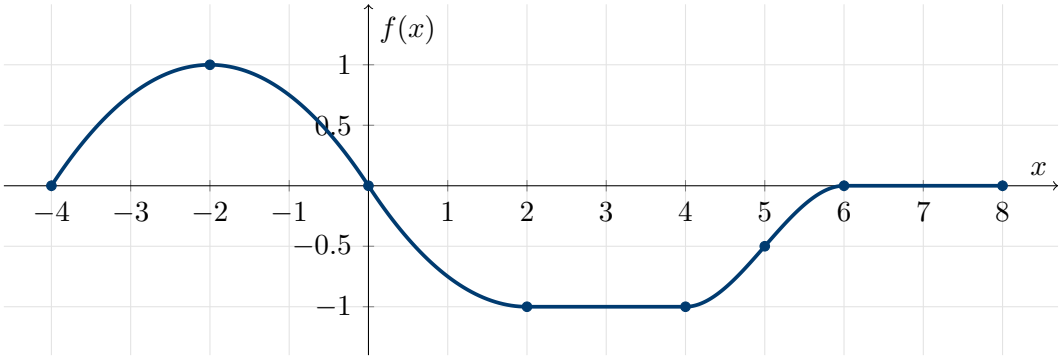


(a) Find the critical numbers of f, and classify each as a local maximum, a local minimum, or neither.

(b) Determine where f is concave up or concave down. Give the x-coordinates of its inflection points.

(c) Sketch f'' wherever it exists, and state where it does not exist in $(-4, 6)$.

7. The graph below shows a function f on $[-4, 8]$. Answer the following questions on $(-4, 8)$.



(a) Where is f increasing or decreasing? Give all x for which $f(x) = 0$.

(b) Where is f concave up or concave down?

(c) For **the derivative** f', give the intervals where it is increasing or decreasing and all x where it equals zero.